Countries can come together to collaborate and thereby accelerate the path going forward in smart manufacturing. For example, this year, the Department of Science and Technology (DST) launched the India-Russia joint technology assessment and accelerated commercialisation programme. Such partnerships aim to help scientific and production houses, researchers, and entrepreneurs in solving the existing challenges and create globally competitive products.

How to implement smart manufacturing

According to Capgemini's Digital Transformation Institute, some manufacturers hired consulting firms to help build a business case and roadmap, others partnered with technology providers for feasibility studies, while the rest used end-to-end technology solutions to start out with smart manufacturing.

A calculated approach involving minimum risks is necessary. A.K. Rao, AGM, NMTronics India, says, "Smart manufacturing should be implemented in a phased manner. The company should define exactly what they require. It is necessary to prepare a roadmap for present and future systems instead of taking 'jugaad', a common practice in India."

Shrikant Sharadrao Borkar says, "Any party—be it from electronics, mechanical, or any other background—needs to see what's going to change technology in the next ten years. Suppose you need a Wi-Fi network and servers, if not upgradable, we can't choose that. If it is not adopted by people, machinery alone can't do anything. All the HODs and other senior leaders can train the employees. People are eager to learn, so management needs to take care."

If you have limited finances, start from the part which you think is more complicated. It is necessary to know what you want to produce and have everything from the purchase of materials to what is happening in AOI under control, and then to ensure quality.

Roberto Gatti says, "To start smart manufacturing, first have a strong infrastructure to support large amounts of data and complete integration of machines including the Internet, good servers, and software for data collection. If you have no control over what you produce, you will be compelled to accumulate a huge amount of components."

Nikhil Pal says, "First, the management perspective needs to change. Also, we can't just allocate budgets, identify one pilot area, and do the implementation there. After implementation, train the people and share the result with the rest of the organisation. Once you see value in that, gradually deploy to other areas as well. Don't start with heavy investment. For example, we can use an IoT gateway to take data out from existing equipment and have our backend analyse the data. Look at how much is the reduction in downtime and quality defects, improvement in productivity, more products you can manufacture, customer satisfaction, and market share. Convert these into numbers when going for ROI calculation."

What SMEs can do

According to a 2018 Smart Manufacturing report, manufacturers included in 'SME's Manufacturing in the New Industry 4.0 Era' survey said that the top barriers preventing smart technology were lack of corporate leadership to lead or plan a smart manufacturing strategy and lack of skill set to manage implementation.

For SMEs to start implementing smart manufacturing of electronics facilities keeping finances into consideration, they need to look for possible financing, renting options, and sharing options for smart manufacturing to work. Manufacturers can aid this progress by taking the responsibility to educate their consumers about the different models based on their budget, capacity, and speed requirements so that they know what to buy. Various local solutions are available for assembly, testing, and more. SMEs can even collaborate with each other for machines. A.K. Rao says, "Different models with capacity and speed are available based on budget. Customers need to check if machines are upgradable. Sometimes they don't know what to buy."

Companies need to optimise their process before installing smart equipment, if they don't have enough budget. SMEs can use existing equipment with modifications. For example, adding an IoT gateway for data processing and analysis. They can also look for renting local solutions under Atmanirbhar Bharat. Once they get revenue, they can purchase the solution. Manufacturers and sellers need to be willing in terms of financing options like accepting payments in instalments.

What happens to jobs?

Many people fear that new techniques like AI in AOI can lead to the loss of their jobs. But contrary to beliefs, even the employees can benefit from smart manufacturing. It can provide more jobs because the technologies need to be managed, controlled, maintained, and developed.

Technology is always focused on reducing repetitive work we can automate. For example, before AOI, operators used to do tedious jobs like inspecting solder joints, while now we can tell if it's a true or false failure using pictures. People can focus on higher value-added jobs that require complex problem solving, critical thinking, creativity, which machines are not capable of right now.

Roberto Gatti says, "Manufacturing to a higher level is not removing jobs but creating more on the production side; the manufacturing process itself may not need more than a few people though. For example, repairing is usually expensive, and correcting huge numbers of errors leads to frustration in workers. Smart manufacturing equipment is helping the manufacturing process to be better and workers to be better."

A.K. Rao says, "People are, in fact, more excited with technology. Humans can't produce or see complex PCB assemblies. There is fatigue from repeatability. The jobs will be transformed and skills can be used on other platforms. The new generation can adapt to Make in India better as they will get an opportunity to work in a different way." EFY

This discussion on smart manufacturing equipment for electronics manufacturing took place during India Technology Week, September 2020. To register as a speaker, exhibitor, or audience at India Technology Week please log on to www.indiatechnologyweek.com